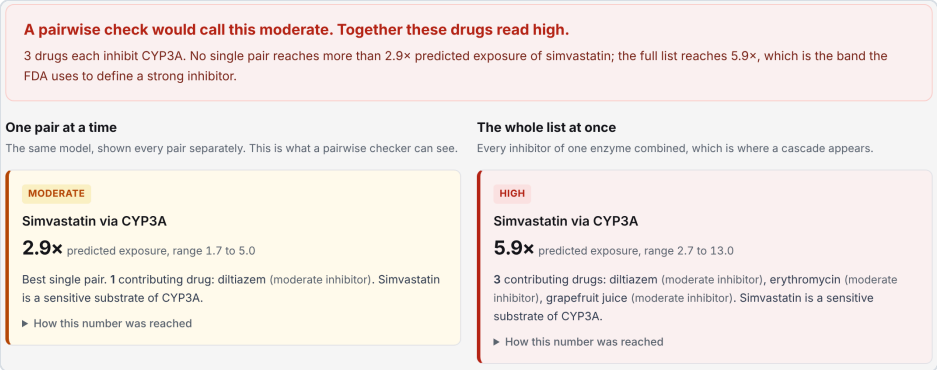
Cascade

The drug interaction a pairwise check cannot see.

Interaction checkers compare drugs two at a time. A patient on seven drugs has 21 pairs, and every one of them can be cleared while three of those drugs each partly inhibit the same enzyme, because the combined effect exists in no pair. Cascade scores the whole list at once, with a model built only from the FDA's published interaction classes, and shows the pairwise view beside it. It takes the list the way people actually write it: brand names, misspellings and pharmacy printouts are read by a small trained model on the server, with no API key.



Simvastatin, diltiazem, erythromycin, grapefruit juice, metoprolol, omeprazole and warfarin. Best single pair: 2.9× predicted exposure, moderate. The three CYP3A inhibitors together: 5.9×, high. Drop any one of them and the finding disappears.

Across the FDA's index table of 205 drugs

Combination	Count	Understated by a pairwise view
2 inhibitors of one enzyme + 1 substrate	84,302	8,057 (9.6%)
3 inhibitors of one enzyme + 1 substrate	1,341,070	78,753 (5.9%)

"Understated": the full list lands in a higher severity band than any single pair of the same drugs. Of random 7-drug lists from the same table, 3.1% hold at least one (62 of 2,000).

How it works

1. **Read**: a neural network trained on realistic misspellings of drug names (RxNorm brands, the 300 most prescribed US drugs) turns a pasted list into table drugs. It asks between look-alikes instead of guessing and never reads a common English word as a drug.
2. **Roles**: 285 published roles for 205 drugs, taken from the FDA's index table. A drug not on it is reported as not covered, never guessed.
3. **Intervals**: every FDA class is a range (a moderate inhibitor raises exposure 2 to 5×), and the ranges are carried through the arithmetic. The fraction metabolised is derived from the class definitions, not assumed.
4. **Combine**: inhibitors of one enzyme add competitively, $A = 1/(1+\Sigma r)$, $AUCR = 1/(fm \cdot A + 1 - fm)$.
5. **Compare**: the baseline is the same model run one pair at a time, so the only difference between the two views is the cascade.

Measured

Every 2- and 3-inhibitor combination in the table	1.4 million enumerated
Calibration: one inhibitor reads back in its own FDA class	holds for all three
Multiplying inhibitions instead, three moderate inhibitors	125× against 13× competitive
Reading 80 blind medication lists, table drugs	precision 99.1%, recall 95.2% (exact-name lookup: 15%)
Look-alike pairs from ISMP's confused-names list	never read as the partner
Tests	62 passing

Limits, stated up front

- It predicts exposure, not harm. Dose, kidney and liver function, genotype and timing are not modelled.
- One enzyme at a time, reversible inhibition only, and only the 205 drugs on the FDA's index list.
- The reader can misread: on the blind lists it added 2 wrong drugs and missed 11, so every reading is shown for checking.
- Not validated against clinical outcomes. It shows what a pairwise view misses, not that a missed cascade harmed anyone.

Educational prototype, not clinical decision support. Data: U.S. FDA, Drug Development and Drug Interactions, Table of Substrates, Inhibitors and Inducers, Tables 1 and 2, retrieved 31 Oct 2026. Names: RxNorm (NLM, Nov 2026); ClinCalc Top 300 of 2024; ISMP List of Confused Drug Names. Built by Joshua Anojulu with Claude (Anthropic), which wrote the code and trained the reader; every number comes from evaluate.py, evaluate_reader.py and the tests in the repository.